

Giridhera Ramanan S

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Portfolio: weirdowrench.github.io | github.com/WeirdoWrench

Profile Summary

AI & Data Science graduate and Full-Stack Developer with hands-on experience building AI-powered web applications using Next.js, FastAPI, PostgreSQL, and modern LLM APIs. Developed and deployed production-style applications including RAG chatbots, computer vision systems, and full-stack SaaS projects. Passionate about backend engineering, AI systems, and building scalable software with clean architecture. Available to join immediately.

Education & Training

Panimalar Engineering College Chennai City Campus, B.Tech Artificial Intelligence & Data Science	Nov 2022 – May 2026 CGPA: 8.22/10
Sitadevi Garodia Hindi Vidyalaya Hr. Sec. Matriculation, Higher Secondary School Education	May 2022 Percentage: 87.3%
Sitadevi Garodia Hindi Vidyalaya Hr. Sec. Matriculation, Secondary School Education	Mar 2020 Percentage: 69.7%

Experience

Java Development Trainee, <i>Robowaves - IN</i>	June 2025 – Aug 2025
- Solved 100+ Java programming exercises covering arrays, strings, collections, object-oriented programming, and exception handling. - Implemented reusable Java programs emphasizing clean coding practices and modular design. - Strengthened problem-solving and debugging skills through practical coding assignments and hands-on exercises.	
IoT Intern, <i>KUBE RESEARCH LABS PRIVATE LIMITED - Chennai, IN</i>	Nov 2024 – Jan 2025
- Developed and integrated BLE-based RSSI tracking and ultrasonic sensor logic with Arduino Uno for an autonomous smart cart, achieving reliable real-time object avoidance within a 30cm detection range. - Designed and executed a 50+ test validation protocol for autonomous cart prototypes, reducing hardware fault rate and driving measurable improvements in navigation stability and sensor response time.	
AI/ML Intern, <i>InternPe - Remote</i>	Aug 2024 – Sep 2024
- Developed five machine learning projects including Breast Cancer Detection, Diabetes Prediction, Email Spam Detection, Car Price Prediction, and IPL Match Prediction using Scikit-learn. - Performed data preprocessing, feature engineering, model training, and evaluation using Python, Pandas, and Scikit-learn. - Built reusable prediction pipelines and documented experiments for model comparison and testing.	

Skills

Programming Languages: Python • Java • JavaScript • TypeScript • SQL • Bash • R (Basic)
Frontend: React.js • Next.js • HTML5 • CSS3 • Tailwind CSS • Vite
Backend: FastAPI • Node.js • REST APIs • JWT Authentication
Databases: PostgreSQL (Neon) • MySQL • SQLite
AI / Machine Learning: TensorFlow • PyTorch • Scikit-learn • OpenCV • MediaPipe • LangChain • NumPy • Pandas • LLMs • Prompt Engineering • Retrieval-Augmented Generation (RAG)
Cloud & Deployment: AWS • Render • Vercel • Netlify
Developer Tools: Git • GitHub • Linux • Postman • Jupyter Notebook • VS Code

Projects

Aura – AI-Powered Mental Health Journal <i>Next.js • FastAPI • PostgreSQL • LangChain • Groq • Together AI • OpenRouter</i>	Live Demo: aura-journal.netlify.app
- Designed and developed a full-stack AI-powered mental health journaling platform using Next.js, FastAPI, and PostgreSQL. - Integrated multiple LLM providers including Groq, Together AI, and OpenRouter with automatic fallback routing for improved service availability. - Implemented secure REST APIs with user authentication, encrypted database storage, regex-based PII masking, and request rate limiting using SlowAPI. - Implemented AI-assisted journal analysis, mood tracking, speech-to-text journaling, anonymous community posting, and interactive dashboard visualizations. - Deployed frontend and backend using Vercel and Render while configuring PostgreSQL on Neon for cloud-based persistence.	

Portfolio AI Chatbot

Live Demo: weirdowrench.github.io

FastAPI • LangChain • Groq • Llama 3.1 • Render

- Built an AI-powered recruiter assistant capable of answering questions about projects, skills, and experience using Retrieval-Augmented Generation (RAG) with resume content as contextual knowledge.
- Developed a FastAPI backend that parses resume PDFs and generates contextual responses using LangChain and Groq's Llama 3.1 model.
- Optimized deployment by leveraging long-context LLM capabilities instead of maintaining an external vector database.
- Embedded a responsive JavaScript chat widget into the personal portfolio with real-time API communication.

Road Accident Detection using MobileNetV2

Github: github.com/WeirdoWrench/RAD

TensorFlow • OpenCV • SMOTE • ImageDataGenerator

- Developed a real-time road accident detection system using TensorFlow, MobileNetV2, and OpenCV.
- Trained the model using the "Car Crash or Collision Prediction Dataset" from Kaggle (~10,000 labeled images), applying preprocessing, augmentation, and class balancing before model training.
- Achieved 86% precision after threshold tuning and model calibration for real-time inference.
- Built a frame-by-frame video prediction pipeline capable of detecting accidents and generating visual alerts.

Posture Monitor – Real-Time Posture Correction System

Github: github.com/WeirdoWrench/posture-monitor

Python • OpenCV • MediaPipe (Pose Landmarker) • Computer Vision

- Built a real-time posture monitoring application using MediaPipe Pose Landmarker with personalized multi-stage calibration for different users.
- Designed adaptive thresholds that combine the measured distance between a user's good and bad posture with a noise floor derived from their own calibration jitter (std-dev), preventing false positives across different webcams and body types.
- Engineered a sustained-drift alert pipeline (8s hold time, 15s cooldown) with desktop + audio notifications via plyer and a terminal-beep fallback, plus a startup self-test so notification failures surface immediately instead of mid-session.
- Added a headless background mode and session analytics (% time in good/bad posture, alert count), and made all file I/O resolve relative to the script's own path to eliminate environment-dependent path bugs.

Moodflix – Movie Discovery App

Live Demo: moodflix04.netlify.app

React 19 • Vite • Tailwind CSS • Appwrite • TMDB API

- Developed a responsive movie discovery platform using React, Vite, Tailwind CSS, and the TMDB API.
- Integrated Appwrite to store search history and generate trending movie recommendations.
- Implemented debounced searching, lazy loading, and graceful error handling to improve performance and user experience.
- Designed a mobile-first responsive interface supporting desktop, tablet, and mobile devices.

Technical Highlights

- Built and deployed 6+ full-stack and AI applications.
- Integrated multiple commercial LLM providers including Groq, Together AI, and OpenRouter.
- Developed REST APIs using FastAPI with PostgreSQL backend.
- Deployed applications on Render, Vercel, and Netlify.
- Experienced with computer vision, NLP, Retrieval-Augmented Generation (RAG), and cloud deployment.

Certifications

Java Programming for Beginners - Simplilearn & SkillUp

AWS Academy Cloud Foundations – AWS Academy Graduate

Fundamentals of Machine Learning & Deep Learning – NPTEL Online Certification

Artificial Intelligence for Real-World Applications – TCS iON

Journey to Cloud: Envisioning Your Solution – IBM SkillsBuild

Introduction to Cybersecurity – Cisco

Languages

English - Full working proficiency

Tamil – Native Proficiency